Retinal Disease Classification: CNN vs. ResNet50 Evaluation

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1 Dataset and Class Distribution

This report details the retinal disease classification project, aiming to classify fundus images into eight distinct categories. Figure 1 displays sample images from each of the eight disease classes.

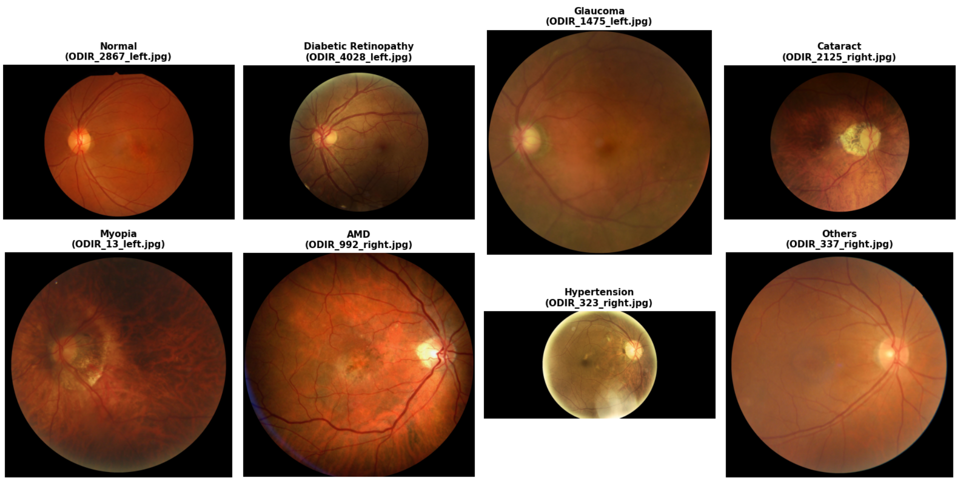

Figure 1: Sample fundus images for the eight retinal disease categories.

An analysis of the dataset revealed a notable class imbalance. The exact sample counts per class are detailed in the class distribution plot shown in Figure 2. To address this imbalance, class weights were applied to the loss function during model training.

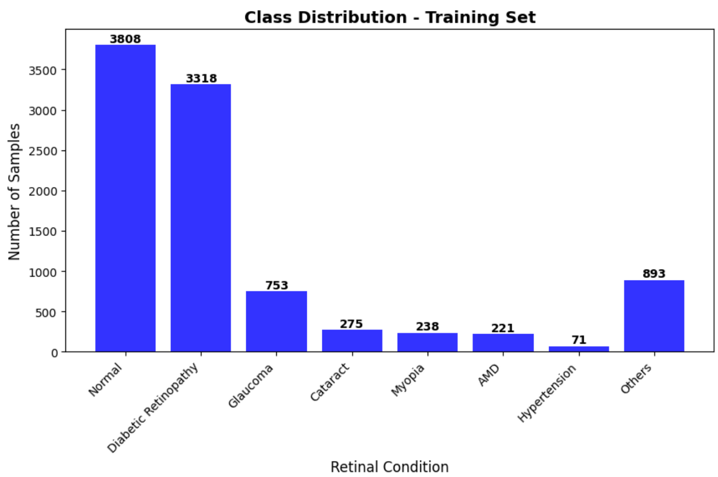

Figure 2: Distribution of samples across the eight disease classes.

2 Custom CNN Training and Evaluation

A Convolutional Neural Network (CNN) was constructed and trained from scratch. Figure 3 shows the training and validation metrics. The plot indicates that the training loss decreases over the epochs, demonstrating reasonable training convergence.

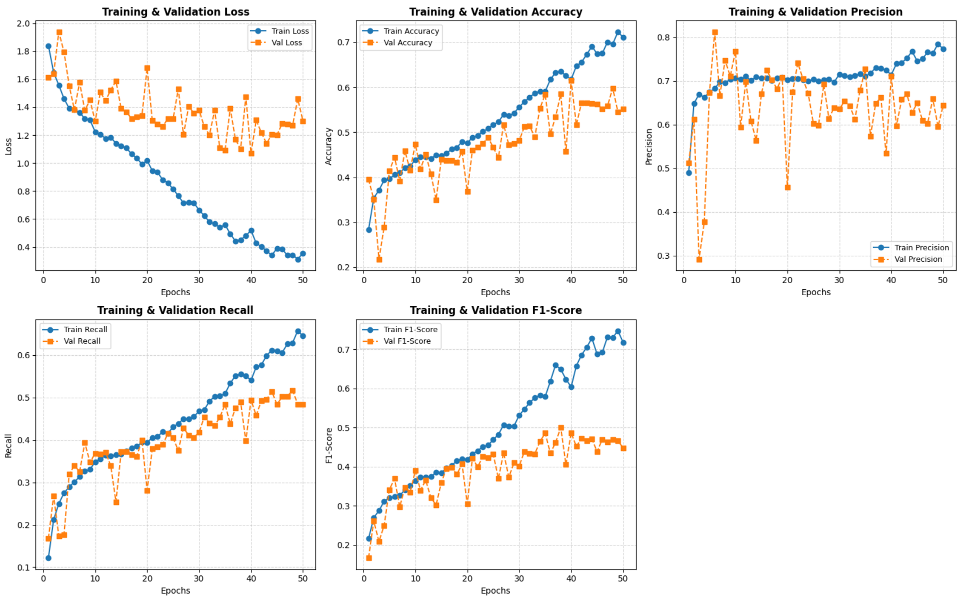

Figure 3: Training and validation loss and accuracy over epochs for the custom CNN.

The model was evaluated on a test set of 1,065 images, achieving an overall accuracy of 0.53. The per-class performance metrics are recorded in Table 1.

Table 1: Per-Class Classification Report (CNN from Scratch)

Class	**Precision**	**Recall**	**F1-Score**	**Support**
Normal	0.78	0.54	0.63	423
Diabetic Retinopathy	0.78	0.53	0.63	369
Glaucoma	0.45	0.69	0.55	84
Cataract	0.42	0.81	0.56	31
Myopia	0.32	0.77	0.45	26
AMD	0.07	0.36	0.12	25
Hypertension	0.25	0.12	0.17	8
Others	0.19	0.28	0.23	99
Accuracy				**0.53**
Macro Avg	0.41	0.51	0.42	1065
Weighted Avg	0.66	0.53	0.57	1065

3 Fine-Tuned ResNet50 Training and Evaluation

A pretrained ResNet50 model was fine-tuned on the same dataset. During the initial setup, it was observed that applying the specific ResNet preprocess function (`tf.keras.applications.resnet50.prepr` is strictly necessary. Without this preprocessing step, the model failed to learn effectively and performed worse than the custom CNN, as illustrated by the stagnant validation curves in Figure 4.

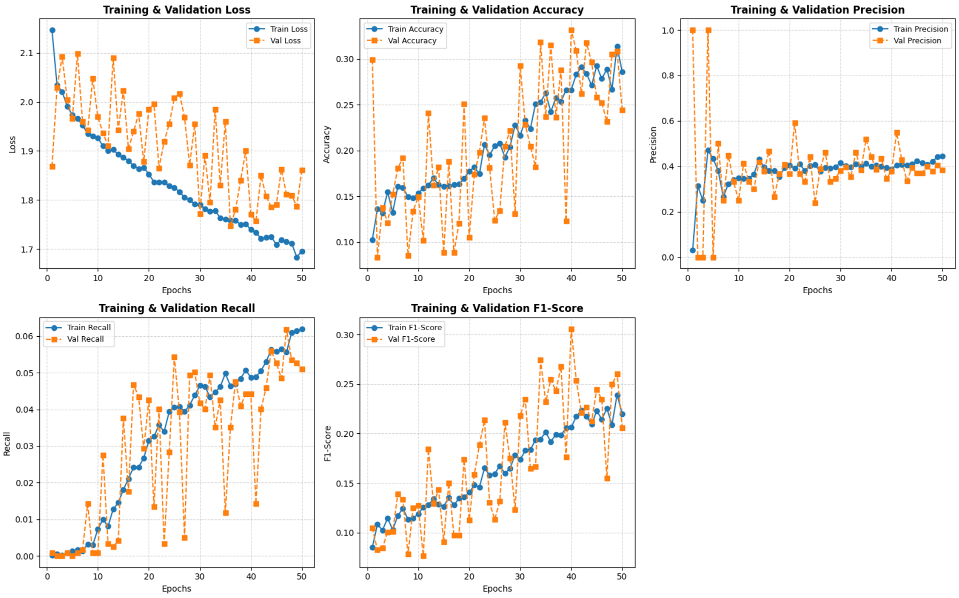

Figure 4: Training metrics displaying poor learning due to incorrect image preprocessing.

Following the application of the correct preprocessing function, the model was trained for 50 epochs. Note that due to an interruption in the display output during the final training run, Figure 5 visually displays only the first epoch's metrics; however, the model completed the full 50-epoch training cycle.

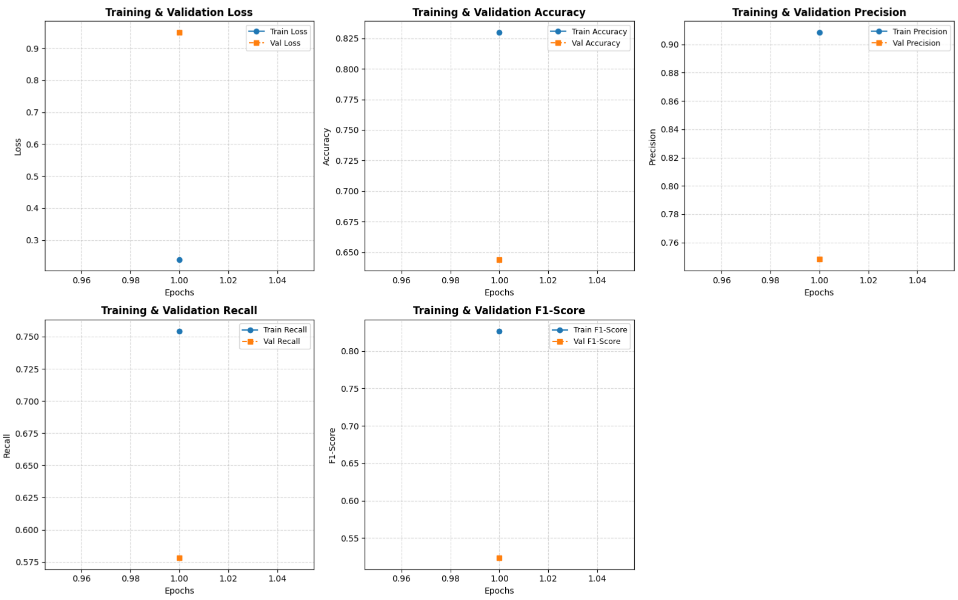

Figure 5: ResNet50 training output (display interrupted after epoch 1; full training comprised 50 epochs).

The ResNet50 model achieved an overall accuracy of 0.67 on the test set. The per-class classification metrics are detailed in Table 2.

Table 2: Per-Class Classification Report (Fine-Tuned ResNet50)

Class	**Precision**	**Recall**	**F1-Score**	**Support**
Normal	0.84	0.66	0.74	423
Diabetic Retinopathy	0.71	0.77	0.74	369
Glaucoma	0.60	0.70	0.65	84
Cataract	0.57	0.87	0.69	31
Myopia	0.63	0.85	0.72	26
AMD	0.35	0.36	0.35	25
Hypertension	0.00	0.00	0.00	8
Others	0.33	0.38	0.36	99
Accuracy				**0.67**
Macro Avg	0.51	0.57	0.53	1065
Weighted Avg	0.70	0.67	0.68	1065

4 Conclusion and Real-Time Deployment

The fine-tuned ResNet50 model outperformed the custom CNN constructed from scratch (0.67 accuracy compared to 0.53) and demonstrated faster convergence under the same number of epochs, provided the images were correctly preprocessed.

Regarding class imbalance, both models struggled with minority classes. For instance, the ResNet50 model failed to correctly classify any instances of Hypertension (support of 8), resulting in an F1-score of 0.00 for that category, while simultaneously achieving high F1-scores for the majority classes like Normal and Diabetic Retinopathy. This indicates that while class weighting helps, severe underrepresentation in training data still restricts the model's predictive capability for those specific diseases.

When considering real-time deployment, there is a trade-off between speed and accuracy. The custom CNN has a lighter architecture, resulting in faster inference times, which is advantageous for low-latency or edge-device applications. However, the ResNet50 model provides a 14% absolute increase in accuracy. Given the medical context of retinal disease classification, where diagnostic accuracy is paramount, the ResNet50 model is the better candidate for deployment, provided the target hardware can support its higher computational requirements. All related project code will be pushed to the designated repository.